

# Postura Equipment Engine

## Complete Developer Reference

Version: May 2026 | Confidential — Postura B.V.

Single reference for the Equipment Recommendations screen. UI changes + backend logic.

### SECTION 1 — UI Changes

Screenshots below show the current screen on the left and the required state on the right.

#### 1.1 Title

**NO CHANGE** — Fixed text. Keep as 'Equipment Recommendations'.

#### 1.2 Subtitle

**CHANGE REQUIRED** — Text change only — reuse existing subtitle Text widget. Replace string with:

Based on your posture analysis and ISO 9241-5:2024 principles...

#### 1.3 Tier message

**CHANGE REQUIRED** — Reuse existing subtitle Text widget — duplicate it, place directly below subtitle. Bind to: EquipmentOutput.tierMessage

| Tier   | String to display                                                    |
|--------|----------------------------------------------------------------------|
| RED    | Immediate ergonomic action required for this workstation.            |
| ORANGE | Plan and implement these equipment changes to reduce ergonomic risk. |
| GREEN  | No urgent actions; maintain current setup and continue monitoring.   |
| YELLOW | Symptoms reported; consider equipment changes and follow-up review.  |

Same style as subtitle — muted colour, no bold. One message per screen only.

#### 1.4 Equipment cards

**NO CHANGE** — Keep existing card layout and sort before rendering (sort order in Section 2, Step 10).

**CHANGE REQUIRED** — Two new fields on every card. Bind as follows:

| Card element | Bind to           | Example                                                             |
|--------------|-------------------|---------------------------------------------------------------------|
| Title        | card.title        | Adjustable monitor riser                                            |
| Description  | card.description  | Height-adjustable stand (8–12 cm) to position monitor at eye level. |
| Source line  | card.sourceLine   | ISO 9241-5:2024; ROSA monitor section.                              |
| Left badge   | card.urgencyLabel | Urgent / Recommended / Preventive                                   |

| Card element     | Bind to             | Example                                                                                                 |
|------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| Right badge      | card.riskBadgeText  | Monitor risk: 3                                                                                         |
| Card note (NEW)  | card.cardNote       | Reuse existing caption/small-text style. Show below description, only when not null.                    |
| Chronicity (NEW) | card.chronicityFlag | Reuse existing warning banner widget (amber). Show text: 'Chronic pattern — prompt action recommended.' |

Left badge — reuse existing pill widgets, text change only:

| urgencyLabel value | Widget to reuse                                                    |
|--------------------|--------------------------------------------------------------------|
| Urgent             | Reuse existing 'High Priority' red pill — text changes to 'Urgent' |
| Recommended        | Reuse existing amber/orange pill — text changes to 'Recommended'   |
| Preventive         | Reuse existing 'Low' green pill — text changes to 'Preventive'     |

Right badge — reuse same existing pill widget, text changes to e.g. 'Monitor risk: 3'.

### 1.5 Bottom buttons

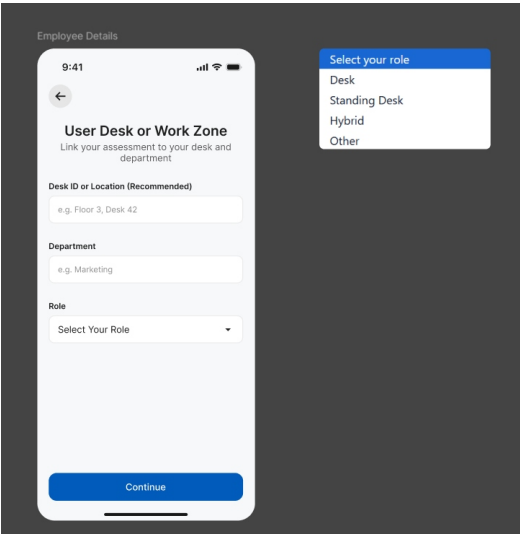
**CHANGE REQUIRED** — Delete 'Download List' and 'Send data to Company' widgets. Keep 'View my results' button exactly as-is — no style change.

| Keep   | Label                | Action                                        | Widget                                           |
|--------|----------------------|-----------------------------------------------|--------------------------------------------------|
| YES    | View my results      | Opens worker's own dashboard (RBAC-protected) | Keep existing full-width blue button — no change |
| REMOVE | Download List        | —                                             | Delete widget                                    |
| REMOVE | Send data to Company | —                                             | Delete widget                                    |

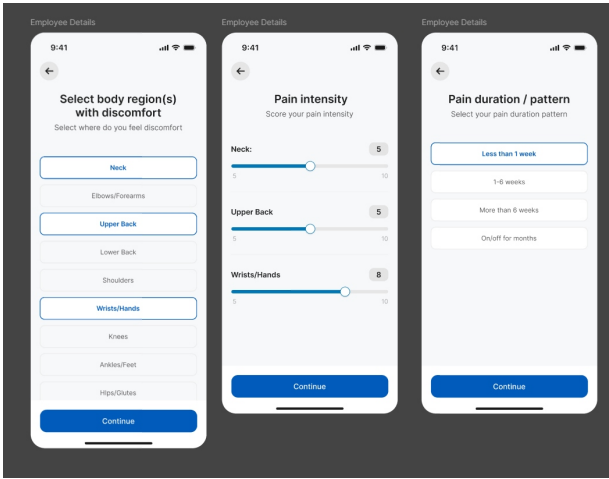
### 1.6 Current UI screens (intake flow)

These are the screens that feed the engine. No layout changes needed to these screens — they are shown here so the developer understands the data flow.

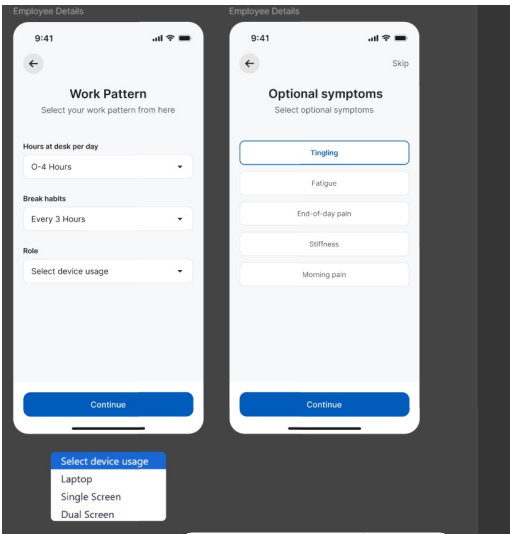
#### Screen 1 — User Desk / Work Zone



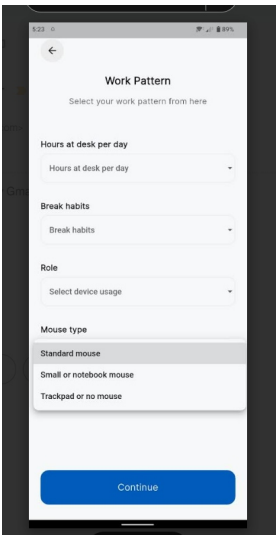
Screens 2–4 — Body regions, Pain intensity (VAS 0–10), Pain duration



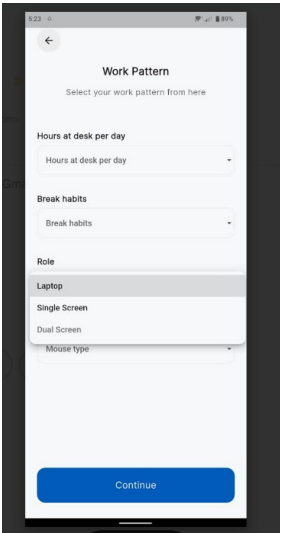
Screen 5 — Work Pattern (device, hours, breaks)



Screen 5 — Mouse type dropdown



Screen 6 — Optional symptoms



SECTION 2 — Backend Engine Logic

Deterministic. No AI call needed for the equipment logic itself. Take inputs, apply rules in order, return structured output.

2.1 Input contract

ROSA sub-scores (from camera + 7 hardware questions):

|                  |      |             |
|------------------|------|-------------|
| rosaChair        | int  | 1-3         |
| rosaMonitor      | int  | 1-3         |
| rosaKeyboard     | int  | 1-3         |
| rosaMouse        | int  | 1-3         |
| rosaChairArmrest | int? | 1-3 or null |

VAS scores — 9 body regions (0–10; null if region not selected on Screen 2):

|              |        |      |
|--------------|--------|------|
| vasNeck      | double | 0-10 |
| vasUpperBack | double | 0-10 |
| vasLowerBack | double | 0-10 |
| vasShoulder  | double | 0-10 |
| vasWrist     | double | 0-10 |
| vasElbow     | double | 0-10 |
| vasKnee      | double | 0-10 |
| vasFeet      | double | 0-10 |
| vasHip       | double | 0-10 |

Tier (from ROSA total score):

|      |        |                                       |
|------|--------|---------------------------------------|
| tier | String | 'GREEN'   'ORANGE'   'RED'   'YELLOW' |
|------|--------|---------------------------------------|

Intake screen inputs (Screens 1, 4, 5, 6):

|                |        |                                                              |
|----------------|--------|--------------------------------------------------------------|
| workZoneType   | String | 'desk'   'standingDesk'   'hybrid'   'other'                 |
| hoursAtDesk    | String | 'h0_4'   'h4_6'   'h6_8'   'h8plus'                          |
| breakHabit     | String | 'every1h'   'every2h'   'every3h'   'rarely'                 |
| deviceSetup    | String | 'laptop'   'singleScreen'   'dualScreen'                     |
| mouseType      | String | 'standard'   'smallNotebook'   'trackpadOrNone'              |
| painDuration   | String | 'ltOneWeek'   'oneToSixWeeks'   'gtSixWeeks'   'onOffMonths' |
| symTingling    | bool   |                                                              |
| symFatigue     | bool   |                                                              |
| symEndOfDay    | bool   |                                                              |
| symStiffness   | bool   |                                                              |
| symMorningPain | bool   |                                                              |

## 2.2 Equipment library — hardcoded constant, do not fetch from API

| ID       | Title                    | ROSA key         | VAS region(s)           | Source          |
|----------|--------------------------|------------------|-------------------------|-----------------|
| equip_01 | Adjustable monitor riser | rosaMonitor      | neck                    | ISO 9241-5:2024 |
| equip_02 | Adjustable footrest      | rosaChair        | lowerBack / knee / feet | ISO 9241-5:2024 |
| equip_03 | Lumbar support cushion   | rosaChair        | lowerBack / hip         | ISO 9241-5:2024 |
| equip_04 | Wrist rest               | rosaKeyboard     | wrist / elbow           | Rempel 2006     |
| equip_05 | Keyboard tray            | rosaKeyboard     | wrist / elbow           | ISO 9241-5:2024 |
| equip_06 | Ergonomic mouse          | rosaMouse        | wrist / shoulder        | Van Eerd 2016   |
| equip_07 | Armrest pads/risers      | rosaChairArmrest | shoulder / upperBack    | Van Eerd 2016   |

## 2.3 Card copy — description + user action per item

| ID       | Description (card body)                                                      | User action                                                                    |
|----------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| equip_01 | Height-adjustable stand (8–12 cm) to position monitor at eye level.          | Raise or lower your monitor until the top edge is at or just below eye level.  |
| equip_02 | Angled footrest with height adjustment (10–15 cm) for stable foot support.   | Add a footrest so both feet are firmly flat and knees sit at ~90°.             |
| equip_03 | Memory-foam lumbar roll to support the lower back in the chair.              | Place lumbar cushion at the curve of your lower back to restore neutral spine. |
| equip_04 | Soft wrist rest to support neutral wrist posture while typing.               | Use a wrist rest to keep wrists neutral when resting between typing.           |
| equip_05 | Under-desk keyboard tray to set keyboard height at elbow level.              | Lower keyboard to at or just below elbow height; keep wrists straight.         |
| equip_06 | Alternative mouse design to reduce lateral wrist and shoulder loading.       | Switch to an ergonomic mouse and keep it directly beside your keyboard.        |
| equip_07 | Adjustable armrest pads (+3 cm or more) to support forearms at elbow height. | Pad or raise armrests to support your elbows at approximately 90°.             |

## 2.4 Step 1 — Which items to include

```
subScore = rosaSubScore for this item
```

```
if subScore >= 2 → include (risk-driven)
if subScore == 1 → include (preventive)
if subScore == 0 → exclude
```

## 2.5 Step 2 — Base urgency

```
vasScore = highest VAS value for this item's linked region(s)
```

```
if subScore >= 2 AND vasScore >= 4 → urgencyLevel = 3 ('Urgent')
if subScore >= 2 → urgencyLevel = 2 ('Recommended')
if subScore == 1 → urgencyLevel = 1 ('Preventive')
```

## 2.6 Step 3 — Exposure multiplier (hoursAtDesk)

```
if hoursAtDesk in ['h6_8', 'h8plus'] AND urgencyLevel == 2
    urgencyLevel = 3 // sustained exposure escalates Recommended → Urgent

if hoursAtDesk == 'h0_4' AND urgencyLevel == 3
    urgencyLevel = 2 // cap – very low exposure
```

## 2.7 Step 4 — Symptom overrides

```
if symTingling AND vasWrist >= 4
    equip_04.urgencyLevel = 3
    equip_06.urgencyLevel = 3
    equip_04.cardNote += 'Tingling may indicate nerve compression – consult a healthcare provider
    if symptoms persist.'

if symStiffness AND vasNeck >= 4
    equip_01.urgencyLevel = 3

if symFatigue AND hoursAtDesk in ['h6_8', 'h8plus']
    // append to all Urgent cards:
    cardNote += 'Fatigue reported – high desk exposure amplifies strain risk.'

if symMorningPain
    // dashboard flag only – no card change
    output.morningPainFlag = true
```

## 2.8 Step 5 — Device setup modifiers

```
if deviceSetup == 'laptop'
    if rosaMonitor < 3 → SUPPRESS equip_01
    if rosaMonitor == 3 → equip_01.cardNote = 'Use a laptop stand + external keyboard.'

if deviceSetup == 'dualScreen'
    equip_01.cardNote = 'Ensure primary screen is centred; secondary at same height.'
    if vasNeck >= 4
        equip_01.cardNote += ' Neck rotation risk detected – consider single primary screen.'

if deviceSetup == 'singleScreen'
    // no modifier
```

## 2.9 Step 6 — Mouse type modifiers

```
if mouseType == 'trackpadOrNone'
    SUPPRESS equip_06
    if rosaMouse >= 2
        equip_04.cardNote += 'Consider adding an external mouse to reduce wrist load.'

if mouseType == 'smallNotebook' AND rosaMouse >= 2
    equip_06.urgencyLevel = 3 // urgency boost regardless of VAS

if mouseType == 'standard'
    // no modifier
```

## 2.10 Step 7 — Work zone modifiers

```
if workZoneType == 'standingDesk'
    SUPPRESS equip_02 (footrest)
    SUPPRESS equip_03 (lumbar support)
    if hoursAtDesk in ['h4_6', 'h6_8', 'h8plus']
        ADD equip_S1 – 'Anti-fatigue mat' – urgencyLevel = 2

if workZoneType == 'hybrid'
    run full seated rules
    all cards: cardNote = 'Recommendations apply to your seated setup.'

if workZoneType == 'desk'
    // no modifier
```

## 2.11 Step 8 — Break habit modifiers

```
if breakHabit == 'rarely'
    all active cards: cardNote += 'Take breaks every 30–60 min to allow posture variation.'

if breakHabit == 'every3h'
    Urgent cards only: cardNote += 'Increase break frequency to reduce sustained load.'
```

## 2.12 Step 9 — Chronicity flag

```
if painDuration in ['gtSixWeeks', 'onOffMonths']
    all cards with urgencyLevel >= 2: chronicityFlag = true
    rationale += 'Chronic pain pattern detected – prompt action recommended.'

if symEndOfDay == true AND painDuration == null
    treat as painDuration = 'oneToSixWeeks'
```

## 2.13 Step 10 — Build risk badge text

```
riskLabelMap = {
    rosaMonitor:      'Monitor risk',
    rosaChair:        'Chair risk',
    rosaKeyboard:     'Keyboard risk',
    rosaMouse:        'Mouse risk',
    rosaChairArmrest: 'Armrest risk',
}
riskBadgeText = riskLabelMap[rosaSourceKey] + ': ' + subScore
// e.g. 'Monitor risk: 3'
```

## 2.14 Step 11 — Sort cards

- 1. urgencyLevel descending (3 → 2 → 1)
- 2. rosaSubScore descending (3 → 2 → 1)
- 3. equipmentId ascending (equip\_01 → equip\_07, stable tie-break)

## 2.15 Step 12 — Output contract

```
// Return to UI:
{
    tier:           String           // 'GREEN' | 'ORANGE' | 'RED' | 'YELLOW'
    tierMessage:    String           // one of the 4 strings in Section 1.3
    equipmentCards: List // sorted
    morningPainFlag: bool           // dashboard signal – no card shown
}

// EquipmentCard fields:
equipmentId      String      'equip_01' ... 'equip_07'
title            String
description       String
sourceLine       String
category         String      'monitor' | 'chair' | 'keyboard' | 'mouse' | 'armrest'
rosaSourceKey    String
rosaSubScore     int         1 | 2 | 3
linkedVasRegion  String?     'neck' | 'lowerBack' | 'wrist' | 'shoulder' | ...
linkedVasScore   double?
urgencyLabel     String      'Urgent' | 'Recommended' | 'Preventive'
urgencyLevel     int         1 | 2 | 3
riskBadgeText    String      e.g. 'Monitor risk: 3'
cardNote         String?     // null if no modifiers apply
chronicityFlag   bool
```

# SECTION 3 — Published Sources

| # | Reference         | Key finding for this engine                                     | URL                                                                                                     |
|---|-------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1 | ISO 9241-5:2024   | Governing standard — workstation layout + postural requirements | <a href="https://www.iso.org/standard/86222.html">https://www.iso.org/standard/86222.html</a>           |
| 2 | Sonne et al. 2012 | ROSA methodology. ICC 0.88–0.91. Action level ≥ 5.              | <a href="https://doi.org/10.1016/j.apergo.2011.03.008">https://doi.org/10.1016/j.apergo.2011.03.008</a> |

| #  | Reference                    | Key finding for this engine                                      | URL                                                                                                                 |
|----|------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 3  | Barros et al. 2022           | ROSA detects intervention. Chair section –2.9 pts.               | <a href="https://doi.org/10.1186/s12891-022-05490-8">https://doi.org/10.1186/s12891-022-05490-8</a>                 |
| 4  | Rempel et al. 2006           | Forearm support RCT n=182. MSD HR=0.49.                          | <a href="https://doi.org/10.1136/oem.2005.022285">https://doi.org/10.1136/oem.2005.022285</a>                       |
| 5  | Van Eerd et al. 2016         | Systematic review 61 studies. Moderate evidence forearm support. | <a href="https://doi.org/10.1136/oemed-2015-102992">https://doi.org/10.1136/oemed-2015-102992</a>                   |
| 6  | Conlon et al. 2008           | Alternative mouse + forearm board RCT n=206.                     | <a href="https://doi.org/10.1136/oem.2006.032243">https://doi.org/10.1136/oem.2006.032243</a>                       |
| 7  | Bruno Garza & Young 2014     | Alternative mouse reduces lateral loading. Scoping review.       | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12089234/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12089234/</a> |
| 8  | Dalager/Sjøgaard et al. 2017 | VAS valid in office worker populations. RCT n=387.               | <a href="https://doi.org/10.1155/2017/7914134">https://doi.org/10.1155/2017/7914134</a>                             |
| 9  | Johns Hopkins / Mahmoud 2025 | Ergonomic + exercise: neck –38%, wrist –37%.                     | <a href="https://pubmed.ncbi.nlm.nih.gov/41307353/">https://pubmed.ncbi.nlm.nih.gov/41307353/</a>                   |
| 10 | J. Clinical Medicine 2025    | Ergonomic interventions OR 0.64 — 36% lower MSD odds.            | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12073017/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12073017/</a> |
| 11 | Frontiers Public Health 2023 | Ergonomic risk management process for offices.                   | <a href="https://doi.org/10.3389/fpubh.2023.1253141">https://doi.org/10.3389/fpubh.2023.1253141</a>                 |
| 12 | OSHA                         | Ergonomics — control hazards.                                    | <a href="https://www.osha.gov/ergonomics/control-hazards">https://www.osha.gov/ergonomics/control-hazards</a>       |

### AI-assisted design rules (not from a single published study)

- ROSA sub-score  $\geq 2$  as equipment trigger threshold
- VAS  $\geq 4$  as urgency override threshold
- hoursAtDesk  $\geq 6$ h as exposure escalation multiplier
- breakHabit = rarely → note appended to all active cards
- mouseType = smallNotebook → urgency boost regardless of VAS
- symTingling → urgency override for equip\_04 and equip\_06
- Laptop suppresses monitor riser unless rosaMonitor = 3
- Standing desk suppresses equip\_02 and equip\_03
- Chronicity flag triggered by painDuration  $\geq$  gtSixWeeks

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